

APM1110-2

MODELING CAMPUS TRAFFIC: A QUEUEING THEORY APPROACH

Student Flow Efficiency at Dual-Guard Security
Checkpoints

SCENARIO

● Gate 2 at Peak Traffic

Campus security checks at Gate 2. This study models student arrival gaps using an exponential and normal distribution to evaluate how effectively a two-guard team handles real-world student arrivals, especially during peak hours.



DATA COLLECTION PROCESS

Ground observations were simulated and recorded tracking 50 consecutive student entries before 9:00 AM – 10:30 AM classes. Using a stopwatch, the researcher logged the exact time gap (in seconds) between consecutive student arrivals at the security gate. Total elapsed tracking time was 688 seconds.

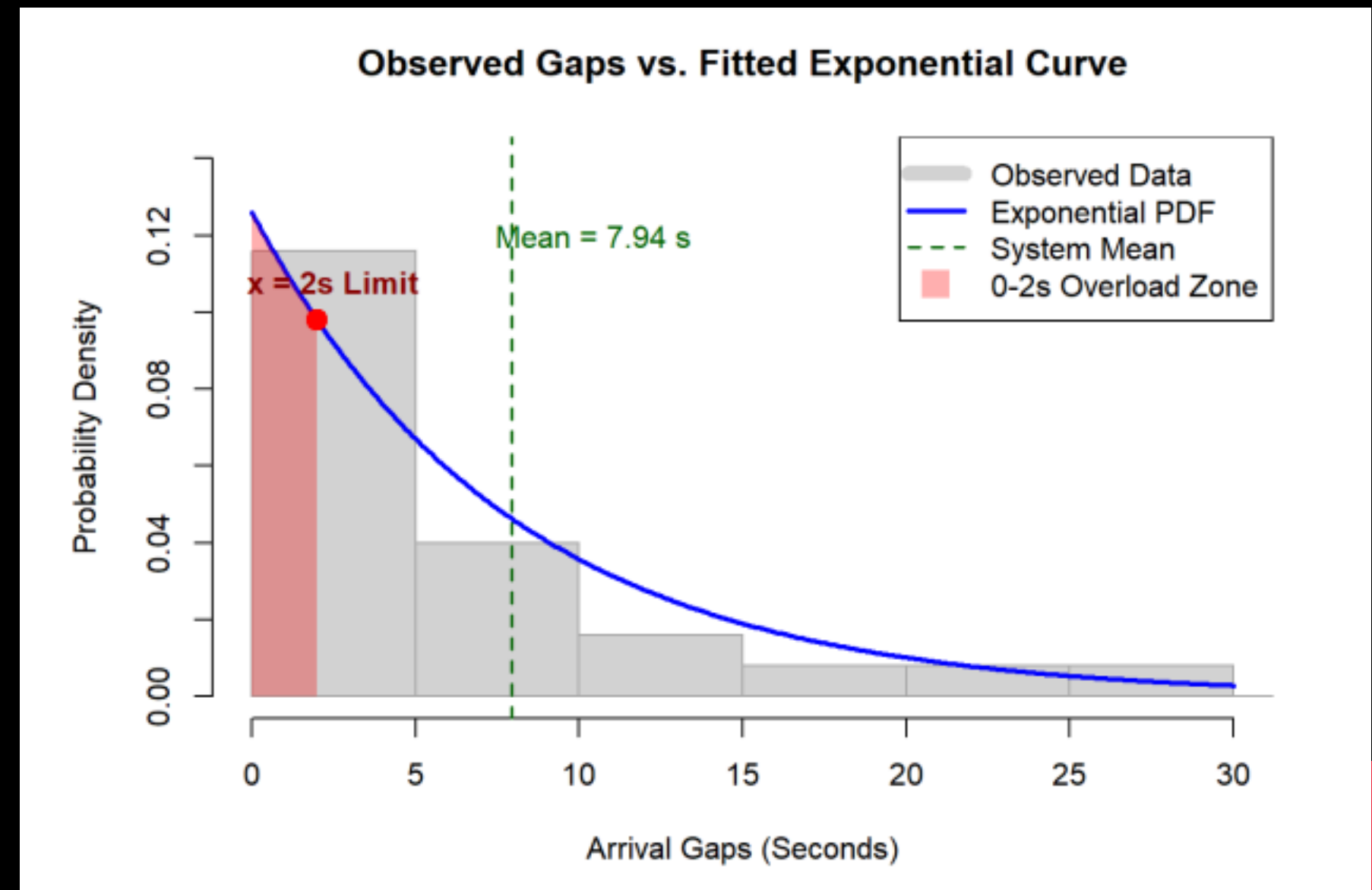
KEY DATA INSIGHTS

- **Arrival Mean: 7.94 seconds**
- **Inspection Time Mean: 5.82**
- **Congestion probability at peak hours: 22.27%**
- **Congestion factor at peak traffic: 2.1**
- **Arrivals at gate per second at peak hours: 0.5**

PROBABILITY DENSITY FUNCTION ANALYSIS

- **Arrivals occur twice as often at peak hours**

Evaluating the PDF reveals that a rapid-fire 2-second arrival gap during peak hours has a density of 0.09790, making it 2.11 times more likely to happen than an arrival at the 7.94-second mean. This proves campus arrival rushes usually occur in bigger batches a few minutes before classes.

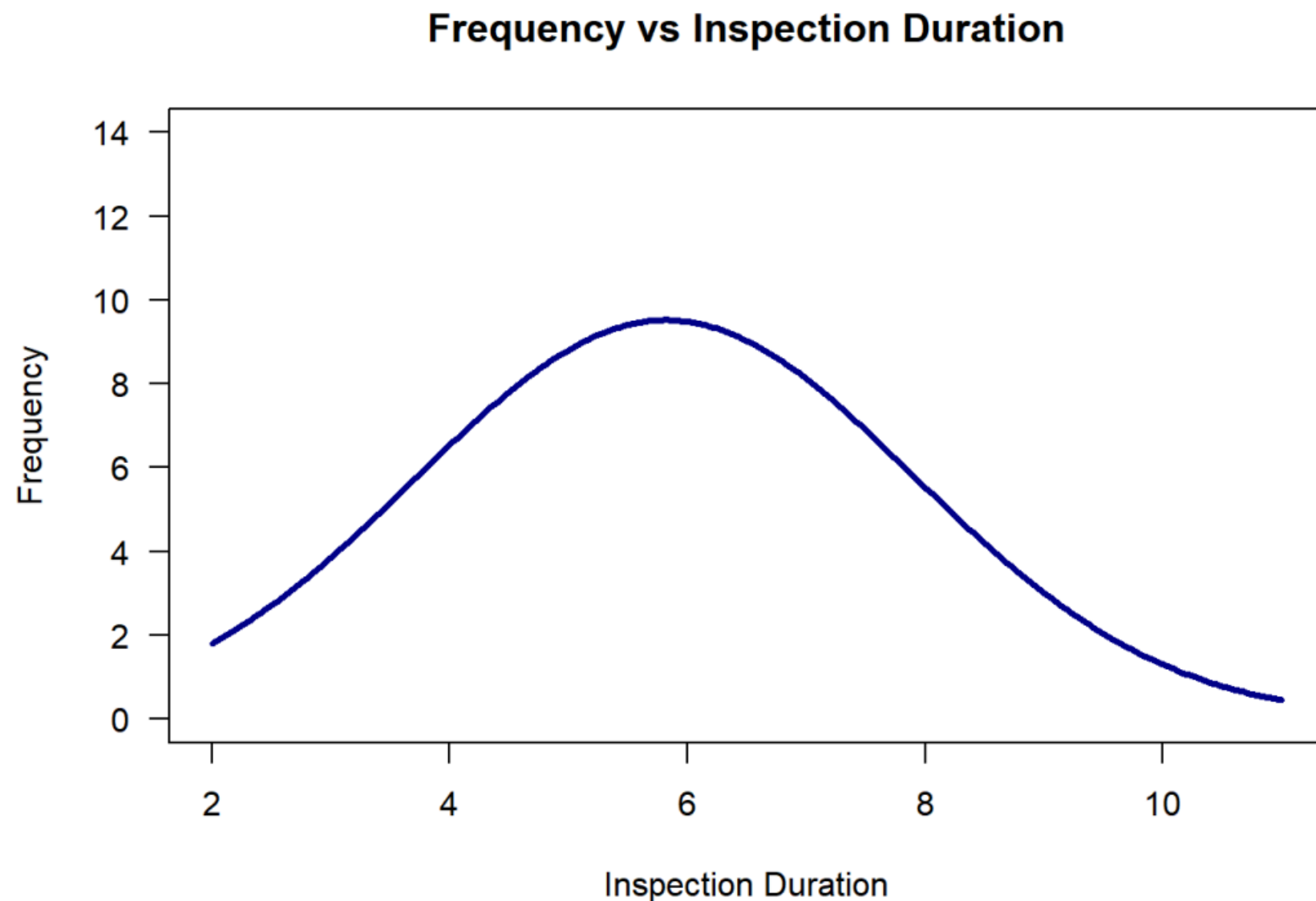


CUMULATIVE DENSITY FUNCTION ANALYSIS

● ~20% chance for congestion, 30 line arrivals
per minute ●

The CDF shows a 22.27% probability that if students arrive within 2 seconds or less an early one-quarter of all traffic saturates the gate during peak hours. At this 2-second saturation point, any gate complication or second lost causes the line to compound at a net rate of 0.50 persons per second, adding 1 arrival every 2 seconds. A minor 30-second stall could generate 15 people onto the line.

NORMAL DISTRIBUTION



● Peak hour arrivals cause congestion rates to rise to 36%

Security checks follow a normal distribution centered at an average (μ) of 5.82 seconds. While the Cumulative Distribution Function (CDF) shows 64% of students clear quickly in 6 seconds or less (with 38% taking 4 seconds or less), a heavy right tail isolates a 36% risk zone where bag complications delay inspections to 7 to 10 seconds. When peak hours hit the gates it runs a deficit against students arriving at said gates. When a guard gets trapped in the 10-second slow zone, the gate degrades to a single lane and capacity is halved.

IMPLICATIONS

For Peak Traffic at Gate 2

2.11

**Congestion
Factor**

Compared to the mean of 7.94 seconds in between arrivals, peak traffic kicks arrivals up to twice the value as traffic arrives at the gates in packs or batches instead of fluidly.

More than a third of all bag checks take 7 to 10 seconds due to human friction (unzipping bags, fumbling for items).

22%

**Arrival
Congestion**

About a fourth of all incoming arrivals lead to line congestion. This means the incoming demand routinely comes through the gate at a pace that saturates and overloads the processing capacity of a two-guard setup.

36%

**Inspections
≥ 7 seconds**

SUGGESTIONS

Instead of keeping a static two-guard setup all day, the university could use a "floating guard" system. Bring in a 3rd guard strictly during the 30-minute peak windows before major class changes to act as a surge protector against the 2.11 congestion shocks.

If a bag check takes longer than 4 seconds, the guard could instruct specific students to step off the line and stand to the side to finish unzipping or searching their bag. The guard can now pull the next student forward, cleanly isolating the 36% slow anomalies from the main traffic stream, preventing delays from turning into a bottleneck.



A dark, moody background featuring a silhouette of a person in profile, looking out of a window at a city skyline. The scene is dimly lit, with the city lights visible through the glass. The overall tone is contemplative and appreciative.

THANK YOU